Lecture 6: Independent Event

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Topics of This Lecture:

1. Independent Event

A B

$P(A|B) = P(A)$

$P(A \cap B) = P(A) \cdot P(B)$

• Event A & B are independent of each other.

1. Independent Event:

If P(A|B) = P(A)
Then we can say A is independent of B.
Occurring of B does not affect the occurring of A.

If P(A|B) = P(A)

Then $\boxed{\dfrac{P(A \cap B)}{P(B)} = P(A)}$

$\boxed{P(A \cap B) = P(A) \,.\, P(B)}$

Then A & B are independent.

If two events A & B are not independent then they are dependent.

Q1. A box contains 4 dairy milk & 2 kit kat,
Raj has taken 2 chocolate & put them back to box.
What is the probability of -
i. 1st pick up is dairy milk then kit kat?
ii. 1st pick up is kit kat then dairy milk?
iii. Getting both dairy milk?

i. (4/6) . (2/6) = (8/36) = (2/9)

ii. (2/6) . (4/6) = (8/36) = (2/9)

iii. (4/6) . (4/6) = (16/36) = (4/9)

$S = \{D\ D\ D\ D\ K\ K\}$ $|S| = 6$

(i) $E_1 = D$ $|E_1| = 4$ $P(E_1) = \frac{4}{6} = \frac{2}{3}$

$E_2 = K$ $|E_2| = 2$ $P(E_2) = \frac{2}{6} = \frac{1}{3}$

$\dfrac{P(E_1) \cdot P(E_2)}{2} = \frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$

(ii) $P(E_1) = \frac{2}{6} = \frac{1}{3}$ $P(E_1) \cdot P(E_2) = \frac{2}{9}$
$P(E_2) = \frac{4}{6} = \frac{2}{3}$

(iii) $E_1 = D$ $P(E_1) = \frac{4}{6}$
$E_2 = D$ $P(E_2) = \frac{4}{6}$
$E_1 \wedge E_2 = \frac{4}{6} \times \frac{4}{6} = \frac{16}{36} = \frac{4}{9}$

Q2. There are three chocolate stores,
Store1 has left 4 dairy milk & 2 kit kat,
Store2 has left 4 dairy milk & 4 kit kat,
Store3 has left 2 dairy milk & 4 kit kat,
You have to buy 1 item from each store.
What is the probability of

a. Getting dairy milk, kit kat & dairy milk from store1, store2 & store3?
b. Getting 2dairy milk & 1kit kat?

S1	S2	S3
4D, 2K	4D, 4K	2D, 4K

$P(D) = \frac{4}{6} = \frac{2}{3}$ $P(K) = \frac{4}{8} = \frac{1}{2}$ $P(D_2) = \frac{2}{6} = \frac{1}{3}$

$\frac{2}{3} \times \frac{1}{2} \times \frac{1}{3} = \frac{2}{18} = \frac{1}{9}$

	S1	S2	S3	
1	D	D	K	$\frac{4}{6} \times \frac{4}{8} \times \frac{4}{6}$
2	D	K	D	$\frac{4}{6} \times \frac{4}{8} \times \frac{2}{6}$
3	K	D	D	$\frac{2}{6} \times \frac{4}{8} \times \frac{2}{6}$

2D 1K

$\frac{4}{6} \times \frac{4}{8} \times \frac{4}{6} + \frac{4}{6} \times \frac{4}{8} \times \frac{2}{6} + \frac{2}{6} \times \frac{4}{8} \times \frac{2}{6}$

$= \frac{2}{9} \times \frac{1}{6} \times \frac{1}{12}$

$= \frac{17}{36}$

1. (4/6) . (4/8) . (2/6) = (1/9)

2. 2 dairy milk & 1 kit kat

dairy milk	dairy milk	kit kat
dairy milk	kit kat	dairy milk
Kit kat	dairy milk	dairy milk

(4/6) . (4/8) . (4/6) + (4/6) . (4/8) . (2/6) + (2/6) . (4/8) . (2/6)
= (2/9) . (1/6) . (1/12)
= (17/36)

Q3. Random Experiment: Tossing two coins
S = { HH, HT, TH, TT }
|S| = 4

E1 = Getting a head on 1st coin
E1 = { HH, HT }
|E1| = 2
P(E1) = 2/4 = 1/2

E2 = Getting a head on 2nd coin
E2 = { HH, TH }
|E2| = 2
P(E2) = 2/4 = 1/2

(E1 ∩ E2) = { HH }
|E1 ∩ E2| = 1
P(E1 ∩ E2) = 1 / 4

If P(E1 ∩ E2) = P(E1) . P(E2) then E1 & E2 are independent
1/4 = 1/2 X 1/2

So E1 & E2 are independent.

Q4. Random Experiment: Rolling two dice,
E1 = Getting a 2 on first die
E2 = Getting a sum of 5
E3 = Getting a sum of 7
What is the probability of

1. E1 & E2 are independent of each other?
2. E1 & E3 are independent of each other?
3. E2 & E3 are independent of each other?

S = { (1,1) (1,2) (1,3) (1,4) (1,5) (1,6)
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6) }

|S| = 36

E1 = Getting a 2 on first die
E1 = { (2,1) (2,2) (2,3) (2,4) (2,5) (2,6) }
|E1| = 6

E2 = Getting a sum of 5
E2 = { (1,4) (2,3) (3,2) (4,1) }
|E2| = 4

E3 = Getting a sum of 7
E3 = { (1,6) (2,5) (3,4) (4,3) (5,2) (6,1) }
|E3| = 6

1. E1 ∩ E2 = { (2,3) }
   If P(E1 ∩ E2) = P(E1) . P(E2) then E1 & E2 are independent
   (1 / 36) = (6 / 36) . (4 / 36)
   = (1 / 36) = (1 / 6) . (1 / 9)
   = (1 / 36) ≠ (1 / 54)

   So E1 & E2 are not independent.

2. E1 ∩ E3 = { (2,5) }
   If P(E1 ∩ E3) = P(E1) . P(E3) then E1 & E3 are independent
   (1 / 36) = (6 / 36) . (6 / 36)
   = (1 / 36) = (1 / 6) . (1 / 6)
   = (1 / 36) = (1 / 36)

   So E1 & E3 are independent.

3. E2 ∩ E3 = 0
   If P(E2 ∩ E3) = P(E2) . P(E3) then E2 & E3 are independent
   0 = (4 / 36) . (6 / 36)
   0 ≠ (1 / 9) . (1 / 6)

   So E1 & E2 are not independent.

- If A & B are independent then,

$\bar{A}$ & $\bar{B}$ are independent,
$\bar{A}$ & B are independent,
A & $\bar{B}$ are also independent.

- A is independent of any event

If P(A) = 1
& P(A) = 0

- Mutually exclusive events with non zero probability are always dependent.

  If A & B are mutually exclusive
  A ∩ B = 0

And if $P(A) \neq 0$ & $P(B) \neq 0$
Then $P(A \cap B) \neq P(A) . P(B)$

Then A & B are dependent events.

Independence of Three events:

A, B & C are independent events if

$P(A \cap B) = P(A) . P(B)$
$P(B \cap C) = P(B) . P(C)$
$P(C \cap A) = P(C) . P(A)$
$P(A \cap B \cap C) = P(A) . P(B) . P(C)$

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Q5. E1 & E2 are two events
P(E1) = P(E2)
P(E1 U E2) = 1
E1 & E2 are independent
The value of P(E1) is
(a) 0, (b) 1/2, (c) 1/4, (d) 1

E_1 E_2

- $P(E_1) = P(E_2)$
- $P(E_1 \cup E_2) = 1$
- E_1 & E_2 are independent

$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

Let $P(E_1) = P(E_2) = x$

P(E1) = P(E2) = x
P(E1 U E2) = 1

So, P(E1) + P(E2) - P(E1 ∩ E2) = 1
x + x - P(E1) . P(E2) = 1
2x - x^2 - 1 = 0
x^2 -2x + 1 = 0
(x - 1)^2 = 0
(x - 1) = 0
x = 1

So P(E1) = 1

$P(E_1 \cup E_2) = 1$

$\Rightarrow P(E_1) + P(E_2) - P(E_1 \cap E_2) = 1$

$\Rightarrow P(E_1) + P(E_2) - P(E_1) \cdot P(E_2) = 1$

$\Rightarrow x + x - x \cdot x = 1$

$\Rightarrow -x^2 + 2x - 1 = 0$

$\Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0$

$\Rightarrow x - 1 = 0 \Rightarrow x = 1$

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Q6. Let P(A) = 1, P(B) =1/2
The value of P(A|B) & P(B|A) are
(a) 1/4, 1/2
(b) 1/2, 1/4
(c) 1/2, 1
(d) 1, 1/2

Hence P(A) = 1
Then A & B are independent events

P(A|B) = P(A) = 1
P(B|A) = P(B) = 1/2

$P(A) = 1$ $\quad$ $P(B) = \frac{1}{2}$

- $P(A|B) = ?$
- $P(B|A) = ?$

$P(A) = 1 \rightarrow$ Independent Event

A & B are independent of each other.

- $P(A|B) = P(A) = 1$
- $P(B|A) = P(B) = 1/2$